

Project Design Phase
Problem – Solution Fit Template

Date	11 September 2026
Team ID	NA
Project Name	Metro Ticket Generating System in ServiceNow
Maximum Marks	2 Marks

Problem – Solution Fit Canvas Breakdown

1. Customer Segment(s) [CS]

- Daily urban metro commuters (office workers, university students).
- Occasional travelers, tourists, and transit passengers requiring single or round trips.
- Metro station staff and Automated Fare Collection (AFC) gate operators.

2. Jobs-To-Be-Done / Problems [J&P]

- Jobs-To-Be-Done: Quickly purchase an accurate transit fare and gain fast, frictionless entry through station turnstile gates.
- Problems Faced: Long queues at physical token vending machines, exact change disputes at manual cash counters, lost or torn paper tickets, and static booking web receipts that turnstile scanners cannot read.

3. Triggers [TR]

- Commuter enters an overcrowded metro station concourse during peak morning/evening hours and sees long lines at manual ticket counters.
- Commuter checks metro transit timings on their smartphone while heading to the station and wants to book their entry pass in advance.

4. Emotions: Before / After [EM]

- Before: Frustrated, rushed, and anxious about missing trains due to ticketing bottlenecks; annoyed by manual currency handling.
- After: Relieved, in control, and confident; experiences contactless, instantaneous gate access using a digital pass.

5. Available Solutions [AS]

- Physical Ticket Counters & Token Vending Machines (TVMs): Prone to long wait times, coin jams, maintenance outages, and high paper token litter/waste.
- Closed-Loop Metro Smart Cards: Convenient for frequent daily commuters, but requires a physical plastic card, minimum balance recharges, and upfront security deposits (not ideal for occasional riders).
- Static Online Confirmations: Email receipts or static SMS confirmations that lack 2D barcode turnstile scannability.

6. Customer Constraints [CC]

- Limited time (must catch trains within tight arrival intervals).
- Unreliable mobile connectivity in underground station corridors.

- Lack of physical cash or exact change (prefer UPI/QR/cards).
- Unwillingness to download heavy, proprietary single-purpose mobile apps for occasional transit.

7. Behaviour [BE]

- Pulls out smartphone while approaching the station entrance.
- Looks for digital signage, QR posters, or web portal booking links to bypass the ticketing booth.
- Relies on mobile payment apps (UPI, Wallets, Cards) for quick contactless payments.

8. Channels of Behaviour [CH]

- 8.1 Online: Mobile web browsers, ServiceNow Service Portal (/sp), responsive web apps, digital UPI payment interfaces.
- 8.2 Offline: Station entry banners, concourse signage, optical Automated Fare Collection (AFC) turnstile barcode scanners.

9. Problem Root Cause [RC]

- Stations rely on manual, centralized physical token dispensing infrastructure that cannot scale during peak surge hours.
- Legacy digital ticketing portals generate standard text confirmations or static PDF invoices without generating real-time, high-contrast, turnstile-scannable 2D barcode tokens.

10. Proposed Solution [SL]

- A self-service ServiceNow Service Portal catalog item (Book A Metro Ticket) that automatically computes single and return journey fares using an onChange client script based on station route matrices.
- Employs an onSubmit script with an interactive modal popup (spModal) using a custom widget (metro_qr_widget) to dynamically generate an encoded 250x250 scannable QR ticket token for direct optical gate access.

Canvas Visual Representation

Problem – Solution Fit Canvas: Metro Ticket Generating System

Define CS, fit into CC	1. CUSTOMER SEGMENT(S) CS - Daily urban metro commuters (office workers & students) - Occasional city travelers, tourists & transit riders - Metro station gate operations & turnstile controllers	6. CUSTOMER CONSTRAINTS CC - Severe time limits (must catch trains within minutes) - Lack of cash or exact currency change at counters - Unwilling to install heavy single-purpose transit apps - Intermittent network connectivity inside stations	5. AVAILABLE SOLUTIONS AS - Physical Token Counters: Massive queues & paper waste - Token Vending Machines: Prone to coin jams & outages - Metro Smart Cards: Requires deposit & physical card - Static Web Receipts: Unscannable at automated gates	Explore AS, differentiate	
	2. JOBS-TO-BE-DONE / PROBLEMS J&P - JTBD: Fast purchase of correct fare & seamless gate entry - Problems: 15–20 min queue times during rush hour - Paper tokens are fragile, easily lost, and non-recyclable - Manual rate calculations cause ticketing errors	9. PROBLEM ROOT CAUSE RC - Over-reliance on centralized, physical ticket booths - Absence of client-side dynamic fare computation scripts - Disconnect between web receipts and turnstile 2D scanners	7. BEHAVIOUR BE - Pulls out smartphone upon entering station concourses - Prefers paying via cashless UPI / QR scanning apps - Walks straight toward automated turnstiles with mobile ready		Focus on J&P, understand RC
	3. TRIGGERS TR - Seeing long ticket counter queues during peak hours - Informational station banners promoting online portal booking	10. YOUR SOLUTION (ServiceNow) SL ServiceNow Service Portal (/sp) Booking Self-service catalog item 'Book A Metro Ticket' Real-Time Dynamic Fare Matrix Client script automates Single & Return rate calculation spModal & Custom Widget (metro qr widget) onSubmit script generates high-contrast 250x250 QR token Master Station Registry & ACL Rules u_metro_station_details with secure role authorization	8. CHANNELS OF BEHAVIOUR CH 8.1 ONLINE - Mobile web browsers accessing ServiceNow Service Portal - Instant cashless digital payment gateways (UPI, Cards) - Backend order fulfillment logs (sc_req_item) 8.2 OFFLINE - Station entrance QR code link posters & informational boards - Physical optical barcode readers at AFC gate turnstiles - Metro gate controller consoles for manual overrides		
4. EMOTIONS: BEFORE / AFTER EM - Before: Frustrated, stressed, anxious about missing train - After: Relieved, confident, in control with an instant gate pass					